BRDF Data Collection and Post-Processing for Various Black Paints

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Abstract. BRDF helps characterize how surfaces scatter light. There is a BRDF measurement system housed in the Meinel building. This machine was used to measure the BRDF data of two black paints. The data produced by the machine was corrected, spline fit, averaged, and plotted, before being prepared for import into FRED.

Keywords: BSDF, BRDF, black paints, scatter, FRED.

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1 Introduction

The bidirectional scattering distribution function (BSDF) is defined as the ratio of the radiance of a scattering surface to the irradiance incident on the surface. The units of BSDF are 1/sr. BSDF is always greater than 0 and is a very weak function of wavelength. Taken from Fest, Equation 1 demonstrates this ratio with a dependence on the elevation and azimuthal angles of the incident ray, θ_i and ϕ_i, and the elevation and azimuthal angles of the scattered ray, θ_s and ϕ_s. The differential radiance of the scattering surface and the differential incident radiance are termed dL and dE, respectively. Figure 1 illustrates the angles used in this definition.

$$BSDF(\theta_i, \phi_i, \theta_s, \phi_s) = \frac{dL(\theta_i, \phi_i, \theta_s, \phi_s)}{dE(\theta_i, \phi_i)} [sr^-1] \tag{1}$$

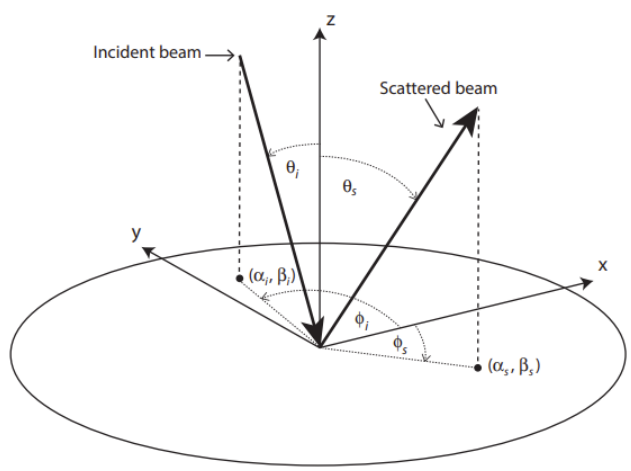

Figure 1: Definition of angles used in Equation 1. Taken from [1]

The scattering surfaces analyzed in this thesis will be small aluminum plates coated in various black paints. As these surfaces are opaque, the type of BSDF recorded will be the bidirectional

reflectance distribution function (BRDF). The assumption will be made that these black paints are isotropic in nature, meaning that the BRDF depends only upon the elevation angle of the scattered ray, not the azimuthal angle. Thus, the BRDF of these paints will only be measured in the plane of the incident beam.

Two paints will be the focus of this thesis: Aeroglaze Z307 (Aeroglaze) and Singularity Black LT (Singularity). Aeroglaze is an industry-standard black paint commonly used in high-quality telescope applications by companies such as Lockheed Martin, Raytheon, and the US Air Force [3]. It is expected to be applied to the majority of the telescope's housing. If the BRDF results of Singularity prove significantly better, it will be used in areas where stray light poses a greater risk to optical performance.

Singularity, however, is composed of carbon nanotubes, which can introduce dust particles into the system. These particles may contaminate optical surfaces, making the paint delicate, difficult to apply, and costly to handle. Despite this drawback, measuring Singularity's BRDF offers an additional benefit, even if the paint is not ultimately used in a future space telescope. Nano-Lab, the company that produces Singularity, has provided its own BRDF data, enabling a direct comparison between their values and those measured in this study. If the results exhibit similar trends, it would confirm that the BRDF machine is properly calibrated and that the Aeroglaze BRDF measurements are reliable. However, because the specific environment and application process used for Nano-Lab's sample are unknown, perfect agreement between datasets is not expected.

2 BRDF Machine

2.1 Machine Layout

The BRDF machine pictured in Figure 2 is an approximately 5x3x3ft chamber containing 3 main components: a stage, a detector, and a laser. The stage is where the sample is placed during data collection and is adjustable along the z axis. The top of the sample must be aligned so that it is at the center of rotation of the BRDF machine which ensures the accuracy of the testing process. The detector is attached to an arm that can rotate about the x-axis. It has a range of motion from -80 to 300 degrees. The laser is attached to an arm that has the freedom to rotate about the z-axis with a range of motion of -90 to 90 degrees about the z-axis. The laser can also be rotated about the center of the stage, ranging from normal to 75 degrees from normal to the sample. A combination of these degrees allows for the measurement of half of a hemisphere for a sample in BRDF mode. It also allows for the gathering of bidirectional transmittance distribution function (BTDF) data for refractive surfaces. It is important to note that the BRDF machine is highly lacking in documentation. Calibration procedures are highlighted, but many of the post-processing had to be derived manually.

2.2 Coordinate System

The program used by this machine is known as RT300s. The coordinate system and terminology of the machine differ compared to Fest's book. The incident angle is defined as the angle between the normal of the measured surface and the laser incident on the surface ranging from 0 to 75 degrees. The azimuth angle is defined as the angle to the arm the laser is attached to is rotated relative to the x-axis ranging from -90 to 90 degrees. The receive angle is the angle at which the detector is positioned relative to the normal of the surface, ranging from -85 to 85 degrees. In RT300s, the incident, azimuthal, and receive angles can be set manually for a single measurement. A sequence can also be defined in which the angles are stepped through as designated and measurements are

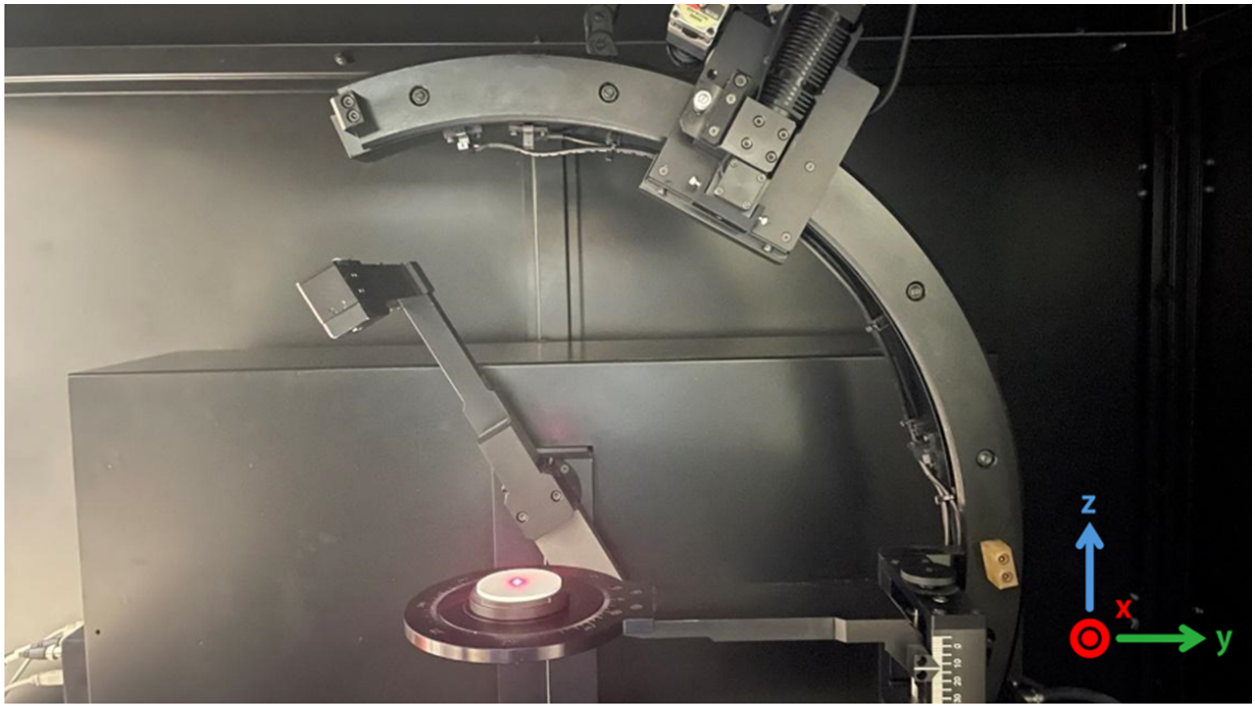

Figure 2: The inner chamber of the BRDF machine with a Lambertian sample placed on the stage.

taken at each step. The sequence used for data collection in this thesis is shown in Section 3.3.

2.3 Calibration

The BRDF machine is designed to measure a wide variety of surfaces, so lasers with various wavelengths and power may be used depending on the surface type. For example, when measuring a plain aluminum sample, a laser with too much power may oversaturate the detector, while a laser with too little power may not adequately measure a sample with a black paint applied. To measure the black samples in this case, a laser was used with a wavelength of 650nm and a spot size diameter of 3.5mm. The BRDF machine must be calibrated for each laser in order to set the value of the intensity incident on the surface. To do this, a Lambertian sample must be used and a calibration file must be created. The Lambertian sample is to be placed on the stage of the BRDF and the stage is to be adjusted so that the top of the sample is central to the arc of the laser and detector. Then, after closing the machine, the incident angle is set to be 8 degrees, the receive angle to 0 degrees, and the azimuth angle to -90 degrees. The reflectance of the sample, .98 for the sample used in this thesis [3], must be entered into the reflectance dialog box. Then, the "Reference Measurement" button is clicked. After the reference measurement is complete, the Lambertian sample must be removed from the machine and the laser covered. Then, after the machine is closed once again, the "Dark Measurement" button is clicked. After this measurement is complete, the "Calibration File Save" button must be pressed and the calibration for the specific laser is saved to its own file. The "Calibration Wizard" dialogue can be seen in Figure 3.

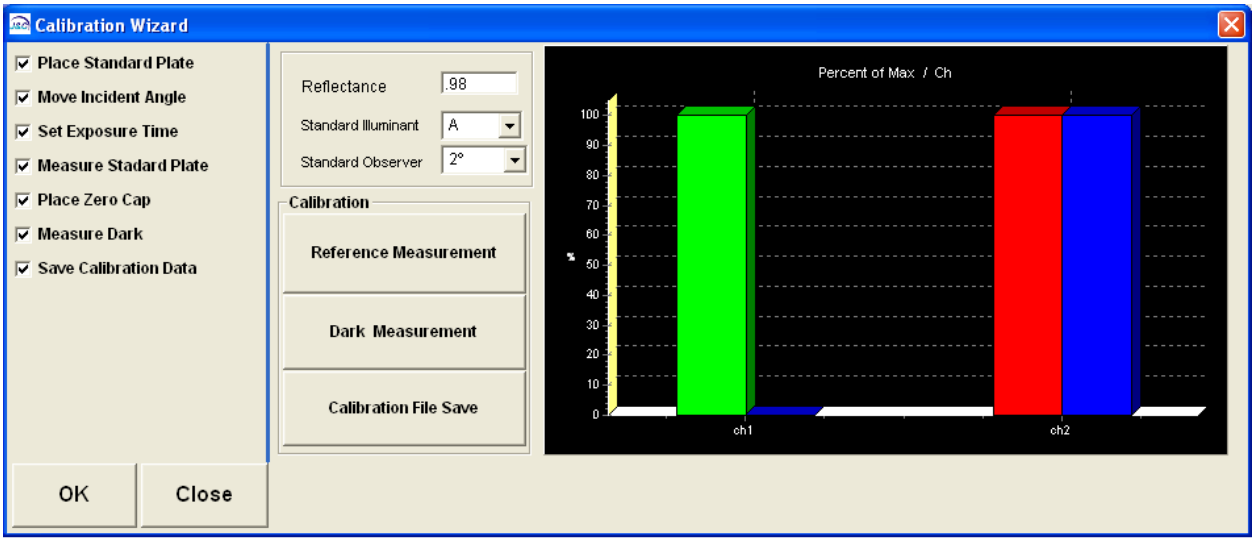

Figure 3: RT300s Calibration Wizard

3 Data Collection

After turning on the BRDF machine and starting RT300s, three steps must be taken before measurements can be made. First, the calibration file for the laser being utilized must be run. Then, the system must be homed. The homing function of the BRDF machine ensures that the arms that hold both the laser and detector are able to move throughout the entirety of their range of motion before returning to their 0 positions. Lastly, the dark level must be set. This function takes a measurement from the detector with the laser positioned directly behind it, allowing no light to reach the detector. This measurement is then set as the base 0 level for all future measurements. It is important to note that this dark level is not always accurate, especially for samples with very low reflectance; this issue came up during data collection and will be addressed in detail in Section 4.2. This issue was present even after recalibrating the machine multiple times and is likely due to some stray light present in the BRDF machine.

After these three steps are complete, a sequence can be set to collect the data. The dialog box prompting the sequence offers the range and step size for the incidence, azimuthal, and receive angles. The program will take a measurement at each designated receive angle for the first incident and azimuthal angle. Then it will rotate to the next azimuthal angle and step through the receive angles once again. Once all azimuthal and receive angles have been stepped through, the incident angle will shift and the process will start again. Once the sequence set has been completed, the RT of each set of angles can be exported to an excel spreadsheet for conversion to BRDF and graphing. A screen capture of the RT300s program with the data spreadsheet can be seen in Figure 4.

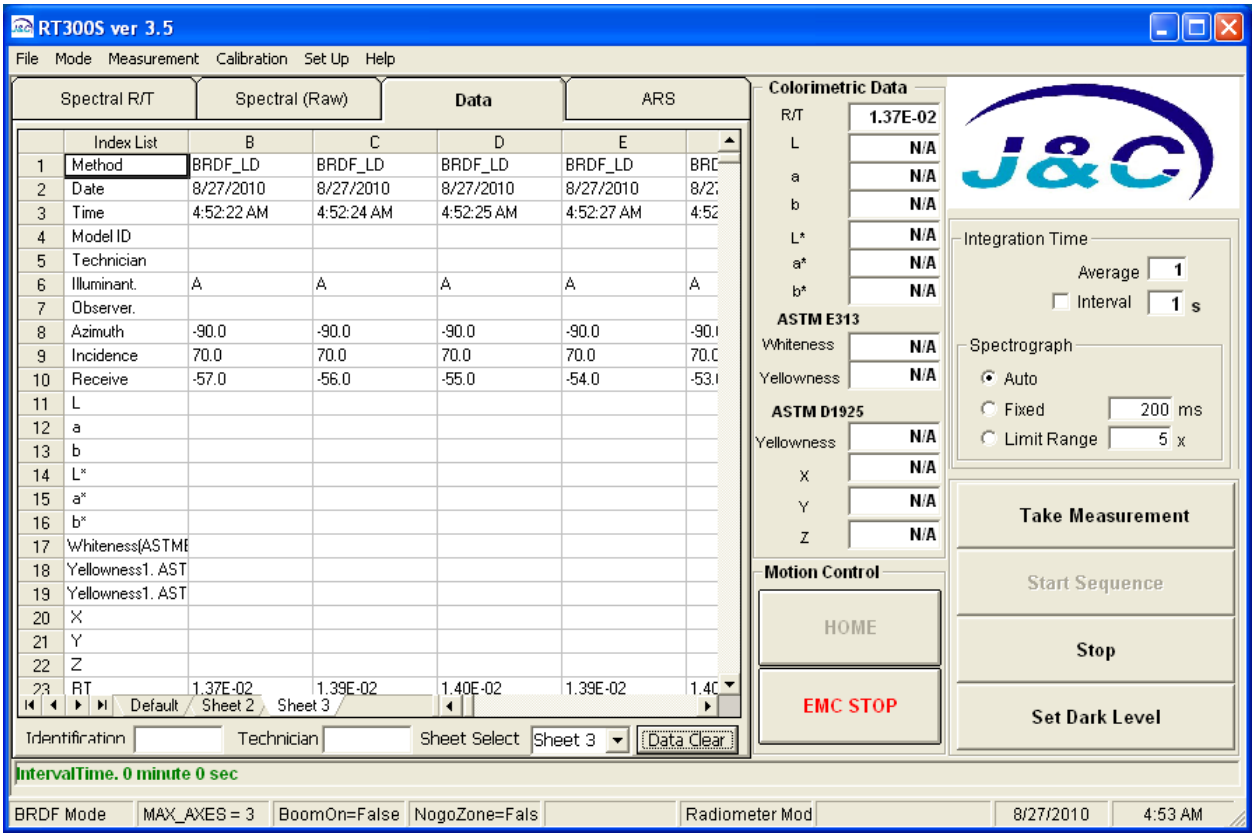

Figure 4: RT300s program with the data spreadsheet

3.1 Machine Data Output

It is important to note that the BRDF machine does not directly measure the BRDF of the samples, but rather the relative reflectance (RT, as denoted by the program) of the sample. RT is unitless and is not explicitly defined in the documentation of the machine. An assumption was made that RT is the ratio of the differential power scattered off the surface at the receive angle ($d\phi_s$)to the differential power incident on the surface ($d\phi_i$). However, this ratio must account for the angle at which the detector is at in reference to the plane of incidence. So the RT value must be divided by a cosine factor. This is shown in Equation 2

$$\frac{RT}{cos\theta_r} = \frac{\phi_s}{\phi_i} \tag{2}$$

Alternatively to Equation 1, the BRDF of the sample can be defined using Equation 3, where $d\Omega$ is the solid angle between the point of incidence and the detector.

$$BRDF = \frac{d\phi_s}{d\phi_i d\Omega_s cos\theta_s}[sr^{-}1] \tag{3}$$

Assuming the scattered light is a hemispherical source the solid angle can be derived for as pi as shown in Equation 4.

$$BRDF = \int_0^{\pi/2}\int_0^{2\pi} sin\phi * cos\theta_s d\phi d\theta_s = \pi[sr] \tag{4}$$

When substituting Equation 2 and 4 into equation 3, Equation 6 is derived, the key equation to findning BRDF values from RT values.

$$BRDF = \frac{RT}{\pi cos\theta_s}[sr^{-}1] \tag{5}$$

$$BRDF = \frac{RT}{\pi \cos(\theta_r)} \text{ sr}^{-1} \tag{6}$$

3.2 Verification of BRDF Equation

This equation can be verified by testing the Lambertian sample used for the calibration of the BRDF machine. Lambertian surfaces should have a constant BRDF [1] independent of incident, azimuthal, and receive angle. So, the Lambertian sample was measured at the incident angles 10, 30, 50, and 70 degrees, azimuth angle -90 degrees, and receive angles -80 through 80 degrees with steps of 1 degree. The RT values of the Lambertian samples can be seen in Figure 5 and the BRDF values can be seen in Figure 6.

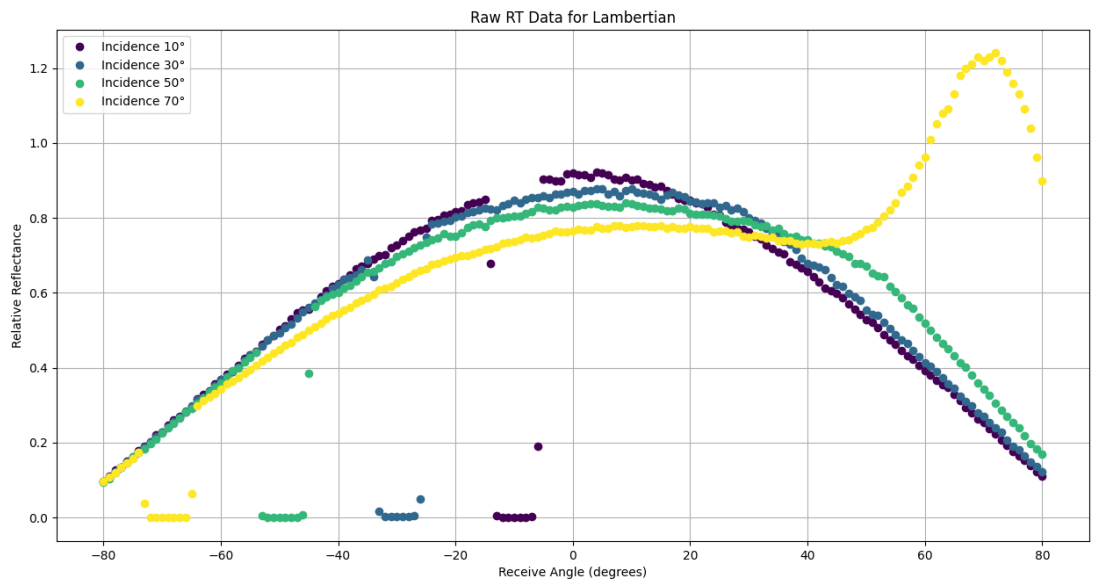

Figure 5: RT Data for Lambertian Sample

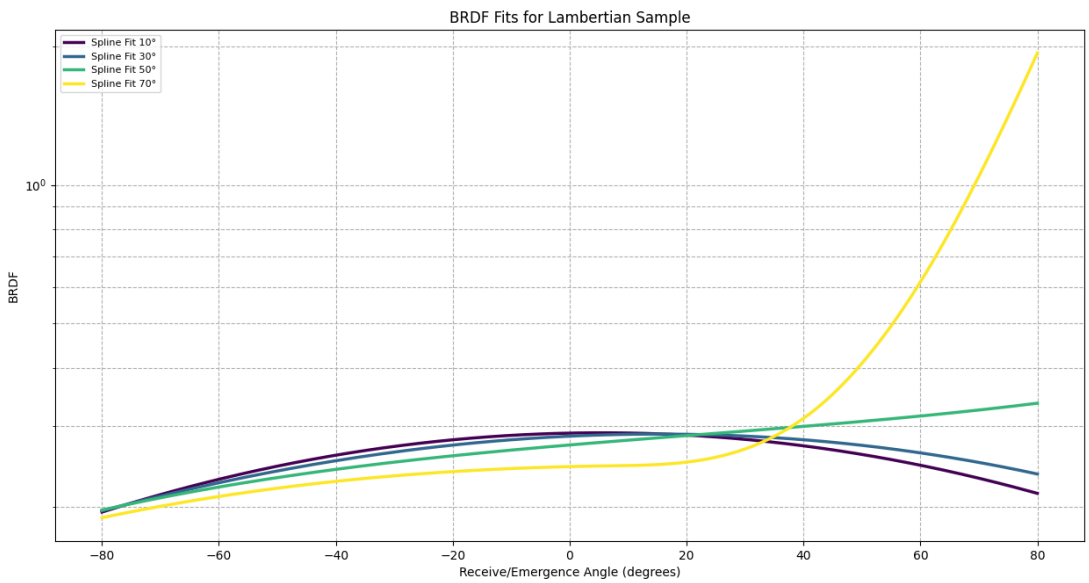

Figure 6: BRDF Data for Lambertian Sample

While for lower incident angles, the BRDF remains nearly constant, for the incident angle of 70 degrees, there is a clear increase in BRDF at higher receive angles. This led to an investigation of the Lambertian sample. As shown in Figure 7, a laser pointer is held at a high incident angle and a paper is held behind the Lambertian sample. There is a clear indication of a specular reflectance at a high receive angle. This means the Lambertian sample is in fact not entirely Lambertian at high incidence angles. However, it is consistent at low incidence angles, so it is still a valid calibration tool. The derived BRDF equation is deemed valid because of this physical investigation and the results at lower incidence angles.

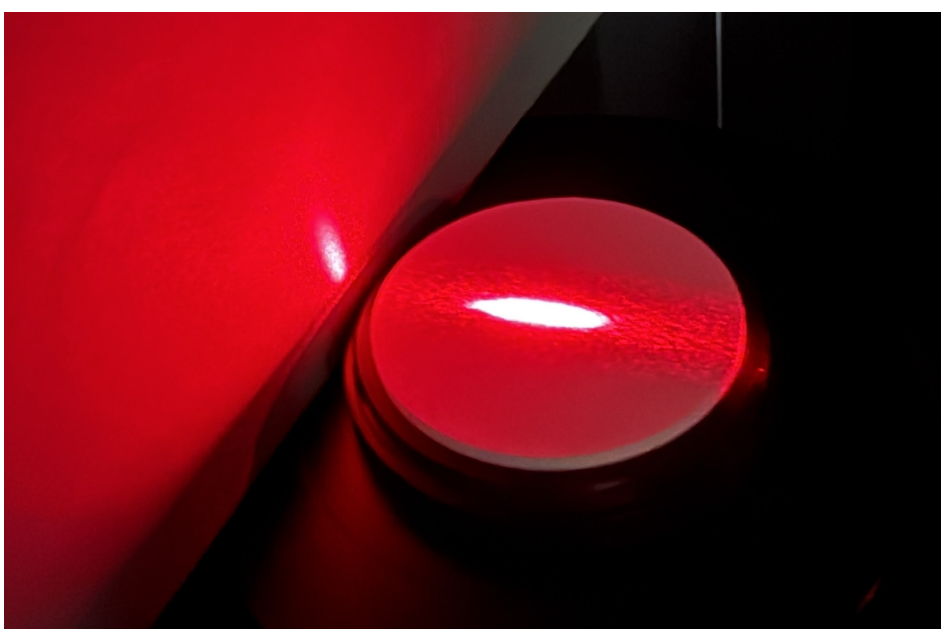

Figure 7: Lambertian sample showing distinct specular reflection.

3.3 Collection Parameters

The collection parameters for Aeroglaze and Singularity will be incident angles 10, 30, 50, and 70 degrees, azimuth angle -90 degrees, and receive angles -80 through 80 degrees with steps of 1 degree. The wide range of incident angles and receive angles allow for minimal interpolation and extrapolation in the programs used for stray light analysis. With the assumption that the black paints are isotropic in nature, the azimuth angle remains constant and all measurements are taken within the plane of incidence. The sequence input in RT300s is shown in Figure 8.

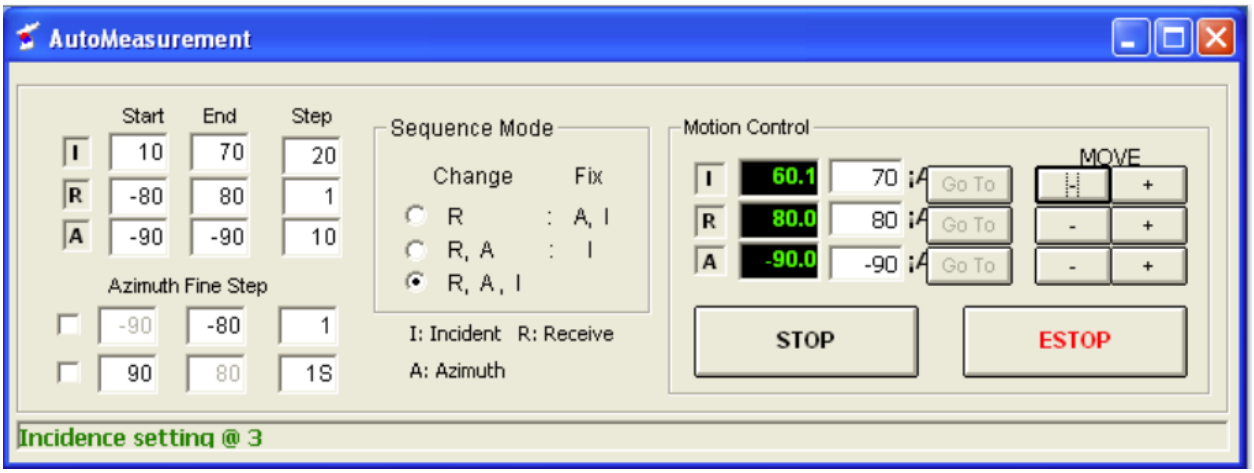

Figure 8: RT Collection Sequence in RT300s

These collection parameters closely match those provided by Nano-Lab. Nano-Lab uses incident angles 10, 30, 55, and 70 degrees, azimuth angle -90, and receive angles -88 through 88 with steps of 1 degree. As the BRDF machine only takes uniform steps in the sequence mode, the incident angle of 50 degrees will be used as a visual substitute for the 55 degree angle used by Nano-Lab.

4 Data Processing

4.1 Plotting Raw Data

Measurements for Aeroglaze and Singularity were taken. Figures 9 and 10 show the raw RT data plotted against the receive angles for each incidence angle. Before converting this data into BRDF, several corrections must be applied to account for factors such as background noise, stray light, and sensor calibration errors.

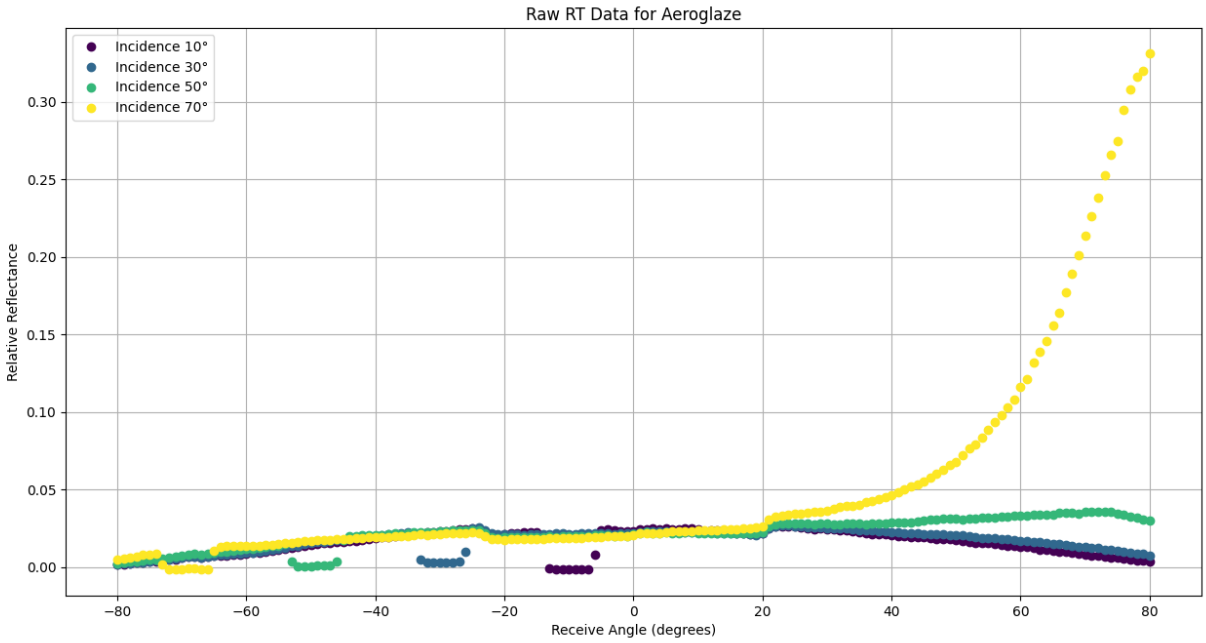

Figure 9: Raw RT Data for Aeroglaze

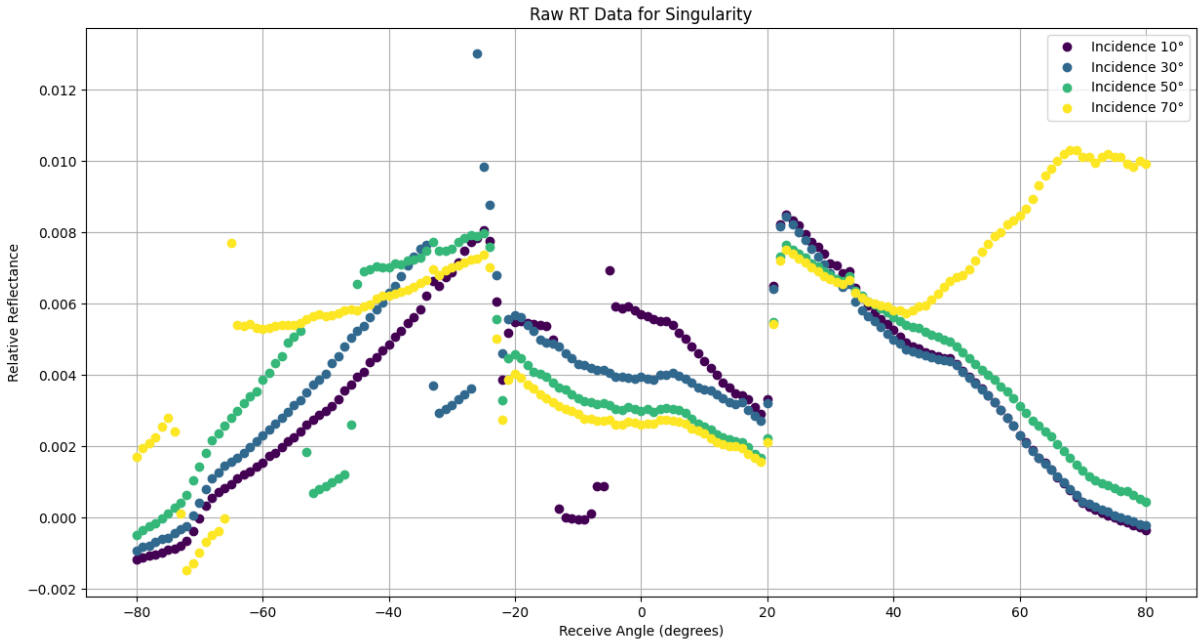

Figure 10: Raw RT Data for Singularity

4.2 RT Data Correction and Conversion

Upon analyzing the plots, three distinct errors were observed and their sources identified. The first source was identified when an error was observed around the receive angle of -25°, where there is a sudden drop in the BRDF data, and at 25°, where a consistent spike appears across all measurements. To investigate this anomaly, a 'blank' measurement was taken using the BRDF machine. The laser was covered and the same sequence was run as when measuring the samples. When plotted, it was found that the detector detected some random RT signal. This "blank" measurement was repeated multiple times with consistent results for the same calibration file. The detected RT was concluded to be due to a consistent error present in the system, potentially stray light within the BRDF machine itself. Thus, this "blank" RT data was subtracted from the RT data of each paint prior to its conversion to BRDF. The reading taken by the detector can be seen in Figure 9.

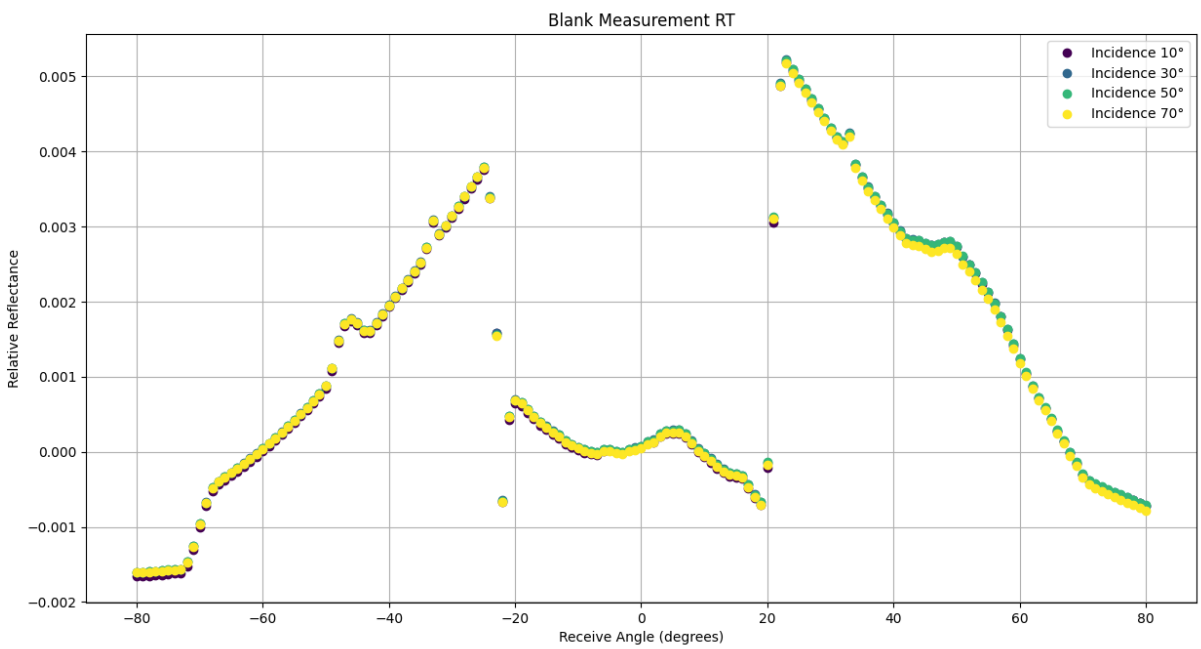

Figure 11: RT detected with laser covered.

The second identified error is the presence of negative RT values in some data points. Since RT values cannot be less than zero, this indicates slight detector error, even after proper calibration. This is likely due to the extremely low magnitude of the measurements. To correct for this, the minimum value (after subtracting the 'blank' measurement) was set to zero as the new floor in the plots.

The third and most apparent source of error is a near zero reading observed when the receive angle is equal to the negative of the incident angle. This is because at this location the detector blocks the laser, meaning no power reflected off the surface is detected. This can be corrected by deleting the data points 5 degrees greater and less than the negative of the incident angle. This is done after the "floor" calibration. The interference caused by the detector is also why the 0 degree incident angle was not measured. Extrapolating a 0 degree incidence would result in an uncertain peak, leading the data to not be fully trustworthy.

After this, the RT readings were converted to BRDF values utilizing Equation 6. The corrected BRDF plots can be seen in Figures 12 and 13.

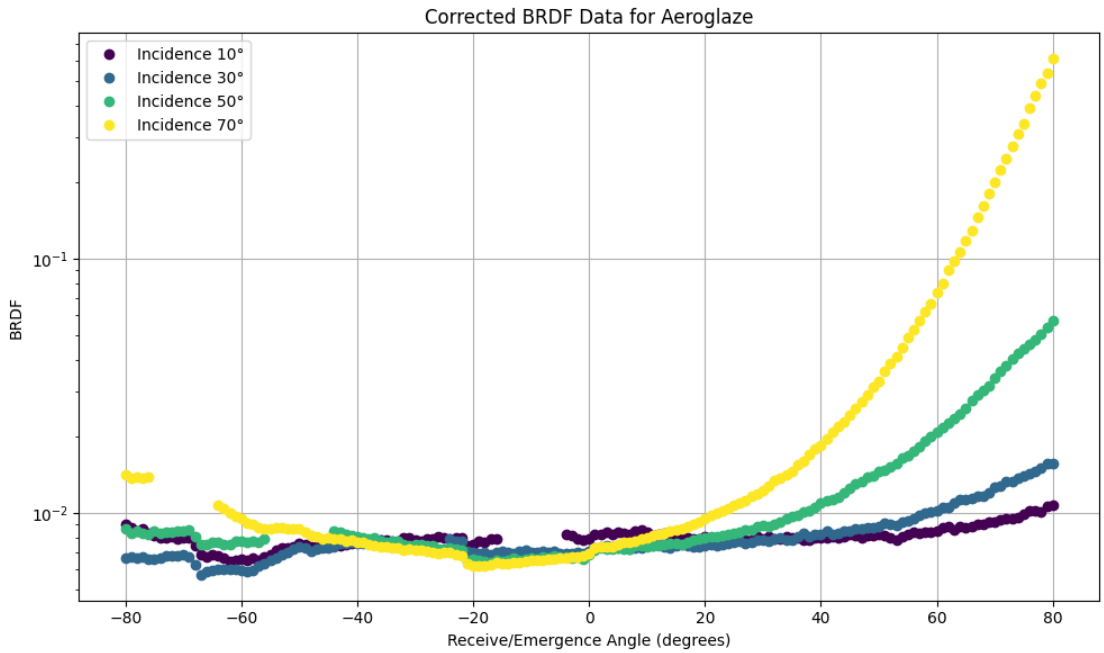

Figure 12: Corrected BRDF Data for Aeroglaze

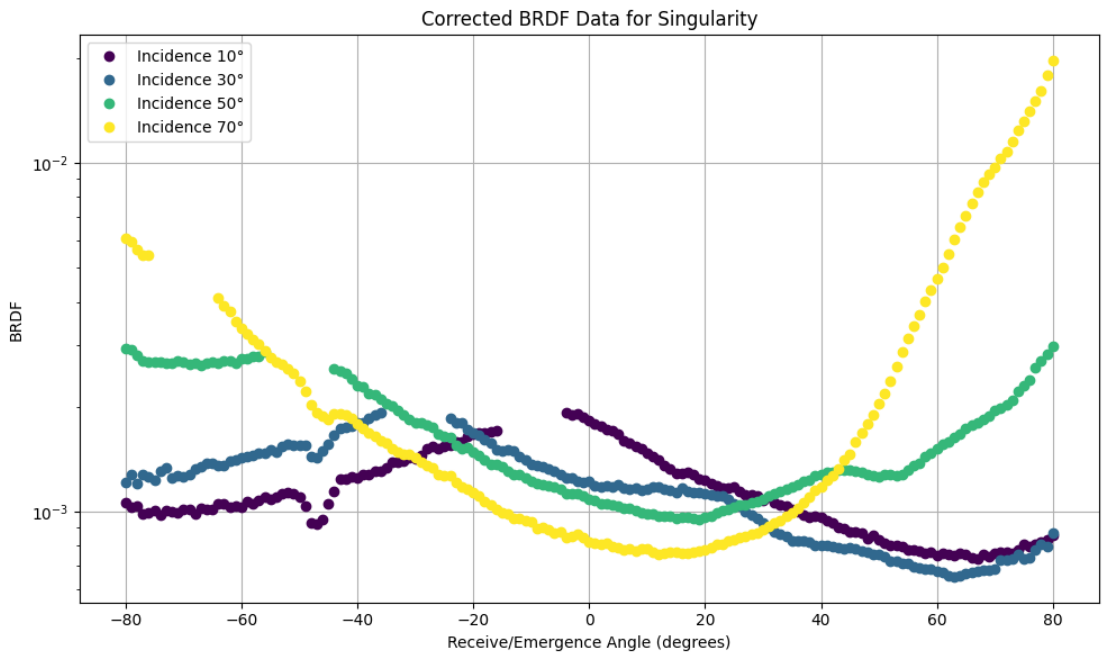

Figure 13: Corrected BRDF Data for Singularity

4.3 Spline Fitting

Once the BRDF data has been plotted and corrected, it is almost ready to be imported into FRED. However, there are still a few discontinuities due to the low amount of power scattered off the surface and missing data points where the detector blocked the laser. So, a log-based spline fit with a smoothing factor of 0.05 is applied to the curves. The paints were measured three times each. Error bars for every other receive angle can be seen in the resulting plots in Figures 14 and 15.

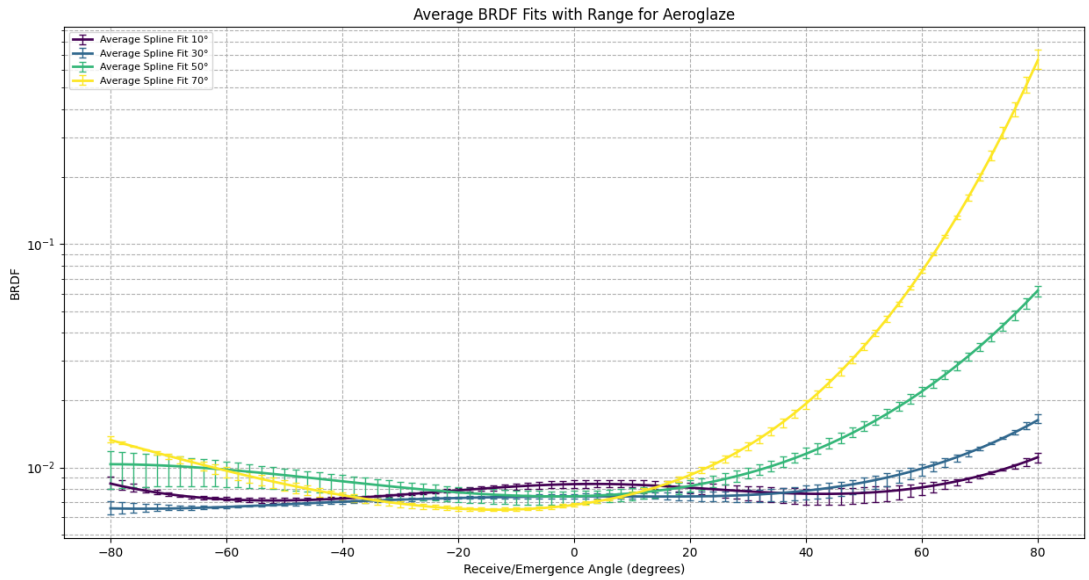

Figure 14: Spline Fit BRDF Data for Aeroglaze

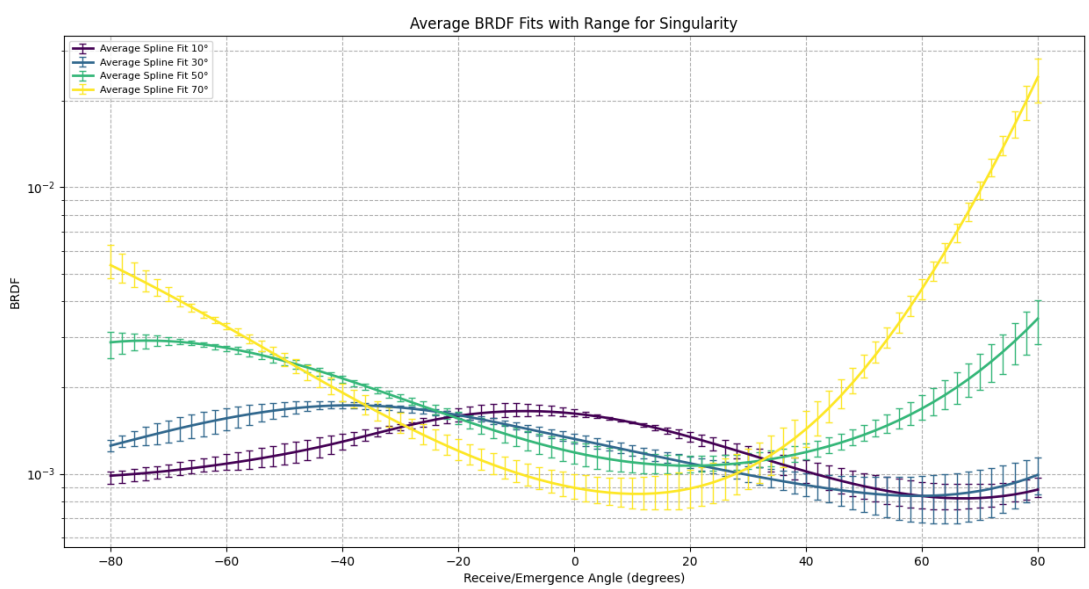

Figure 15: Spline Fit BRDF Data for Singularity

4.4 Singularity Comparison

The measured Singularity data was compared to the BRDF measurements provided by Nano-Lab. However, several factors about Nano-Lab's measurement methods remain unknown, including the measurement environment, the paint application process, and the machine used. These unknowns could contribute to the differences observed between the datasets. While the plots show slight differences, they peak and valley at similar receive angles and exhibit similar trends. Both datasets display a nearly identical peak BRDF for an incident angle of 70 degrees and a receive angle of 80 degrees. Since high incident and receive angles are the most important for BRDF data, this confirms that the measurements from the BRDF machine are valid. The comparison of the data is shown in Figure 16.

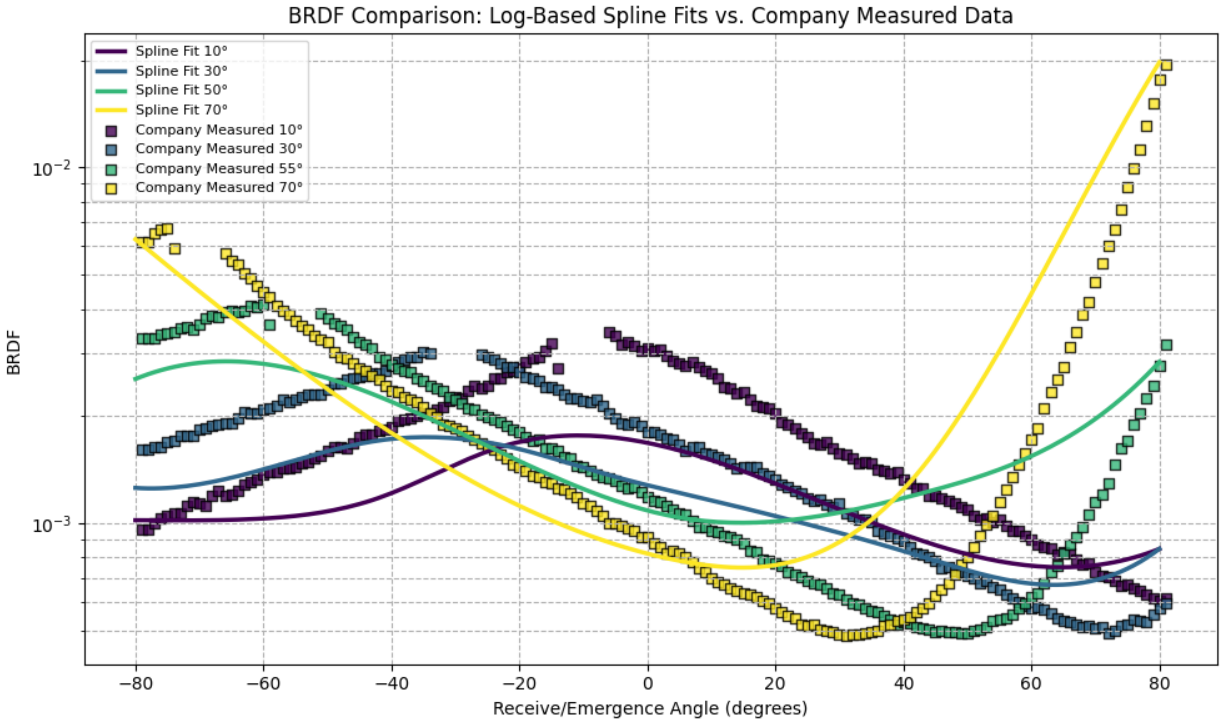

Figure 16: Comparison of Spline Fit BRDF Data for Singularity vs Nano-Lab Measured Data

5 Importing Data into Relative Software

5.1 FRED

Importing this data into FRED is relatively simple. The BRDF data simply needs to be exported into a text file with a specific format. The header is always written as type bsdf_data. The second line is provided as format angles = X bsdf = Y scale = Z. X can be inputted as deg, when the specular and scatter angles are provided in degrees and sin when the angles are provided as direction cosines. Y can be inputted as value when the BSDF is inputted as a normal value, or log when the BSDF value is provided as log base 10 of the value. Z is the scale factor of the BSDF model and is typically set as 1. The header used for the files is shown in Figure 17.

```
type bsdf_data
format angles=deg bsdf=value scale=1.0
```

Figure 17: Header for FRED BSDF text file.

The BRDF values can be inputted for each incident and receive angle combination. The format used by FRED requires the specular and scatter angles for each BRDF value; these angles are equivalent to the incident and receive angles, respectively. The azimuthal angles for both the incident and receive angles are also available for input, but are not necessary for isotropic surfaces such as the black paints being analyzed. The required format is shown in Figure 18.

θ_specular	φ_specular	
θ_scatter	φ_scatter	bsdf
θ_scatter	φ_scatter	bsdf
θ_scatter	φ_scatter	bsdf
Additional scatter data rows as needed		

Figure 18: Format of BRDF text files for FRED.

The BRDF data of Aeroglaze and Singularity was organized into this format. The files were uploaded to FRED and the scatter properties of the paints can now be utilized for stray light analysis.

5.2 ZEMAX

Importing BRDF data into ZEMAX is more complicated than importing into FRED due to differences in coordinate systems. Specifically, the BRDF coordinate system used in ZEMAX differs from that of the BRDF measurement setup and FRED. This section provides a brief overview of these differences, including how to interpret polar plots in ZEMAX. For a detailed guide on importing data into ZEMAX, refer to "Zemax Tabular BSDF Data File Format" [2].

Each BRDF data point in a ZEMAX file requires three angular inputs: the angle of incidence, azimuth angle, and radial angle. The angle of incidence is defined in the same way as in the BRDF measurement system, but the azimuth and radial angles in ZEMAX are referenced relative to the specular reflection direction rather than the surface normal. This angular convention is illustrated in Figure 19, taken from [2].

To adapt the measured BRDF data into ZEMAX format, a Python script was developed. This script first set the azimuth angles to 0° and 180°, splitting the BRDF data accordingly. The radial angles were then defined from 0° to 180°. To match ZEMAX's coordinate system, the data was shifted by the angle of incidence. Any combinations of radial and azimuth angles without corresponding BRDF values were assigned a value of 0.

Each angle of incidence also requires a Total Integrated Scatter (TIS) value. This can be calculated using Equation 7, taken from [1]. θ_s is set to the receive angle from the BRDF machine coordinate system.

$$TIS = \int_0^{2\pi} \int_0^{\pi/2} BSDF \cdot \sin(\theta_s) \cos(\theta_s) \, d\theta_s \, d\phi \tag{7}$$

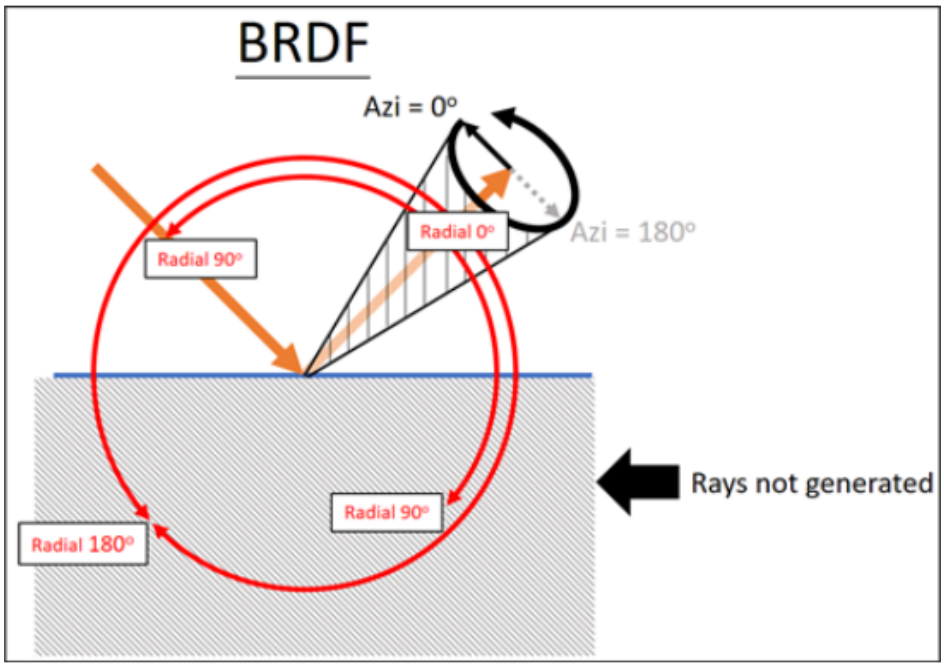

Figure 19: Visualization of radial and azimuthal angles in ZEMAX. Note that these angles are defined with respect to the specular reflection direction rather than the surface normal.Taken from [2]

Overall, converting BRDF data for use in ZEMAX is feasible, but it requires a significantly more complex implementation compared to FRED. Given that FRED is also the superior tool for stray light analysis, it is recommended for future work.

References

[1] Eric Fest. *Stray Light Analysis and Control*. SPIE, Mar. 27, 2013. ISBN: 978-0-8194-9326-2. DOI: 10.1117/3.1000980. URL: https://spiedigitallibrary.org/ebooks/PM/Stray-Light-Analysis-and-Control/eISBN-9780819493262/10.1117/3.1000980 (visited on 04/02/2025).

[2] Ron Rykowski. "Zemax Tabular BSDF data file format". In: *Zemax Knowledgebase* (Mar. 31, 2021). URL: https://support.zemax.com/hc/en-us/articles/1500005486801-Zemax-Tabular-BSDF-data-file-format.

[3] Socomore. *aeroglaze_z307_TDS_english*.